ADMINISTRAÇÃO DE SERVIÇOS DE REDE - ATIVIDADE -01

1) Criação das chaves (pública e privada) no Windows com **OpenSSH** instalado nativamente.

Definido o tipo de criptografia (RSA é o padrão).

Definido o tamanho da chave (4096 bits).

Prompt de Comando - ssh-keygen -t ed25519 -C "sgtgladstone@gmail.com"

```
C:\Users\França>ssh-keygen -t ed25519 -C "sgtgladstone@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\França/.ssh/id_ed25519): C:\Users\França/.ssh/id_Keyed25519
```

Configuração e definição da Senha

```
Enter file in which to save the key (C:\Users\França/.ssh/id_ed25519): C:\Users\França/.ssh/id_Keyed2551
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
```

Confirmação da criação das chaves publica e privada com a identificação do respectivo Diretorio

Prompt de Comando

```
C:\Users\França>ssh-keygen -t ed25519 -C "sgtgladstone@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\França/.ssh/id_ed25519): C:\Users\França/.ssh/id_Keyed25519
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\França/.ssh/id_Keyed25519
Your public key has been saved in C:\Users\França/.ssh/id_Keyed25519.pub
The key fingerprint is:
SHA256:jUFHvyz+k1hzKiN+yjcaFVNhXIgVU0MAqHueK74zofI sgtgladstone@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|        .o+=OO=  |
|       ....+o. . |
|       .. o .    |
|      .  + + .   |
|       .S + o    |
|      o .o .o .  |
|     . +...o =   |
|  . . +.+.B.+    |
|   oE.o**B +..   |
+----[SHA256]-----+
```

Localização das chaves no Diretório

```
C:\Users\França>cd .ssh

C:\Users\França\.ssh>dir
 O volume na unidade C não tem nome.
 O Número de Série do Volume é 8838-B6AD

 Pasta de C:\Users\França\.ssh

11/03/2026  09:30    <DIR>          .
11/03/2026  09:30    <DIR>          ..
03/03/2026  15:13               484 id_ed25519
03/03/2026  15:13               128 id_ed25519.pub
11/03/2026  09:30               464 id_Keyed25519
11/03/2026  09:30               105 id_Keyed25519.pub
               4 arquivo(s)          1.181 bytes
               2 pasta(s)   255.241.793.536 bytes disponíveis
```

Ativar o Windows

Conferido a Chave privada

```
C:\Users\França\.ssh>type id_Keyed25519
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAACmFlczI1Ni1jdHIAAAAGYmNyeXB0AAAAGAAAABD+YdzD3e
Vxt1kfmorSXKz0AAAAGAAAAAEAAAAzAAAAC3NzaC1lZDI1NTE5AAAAIIrD6PQCTK8gwUzd
uvOhDAx0hpa+TNIlxnCYYz9F1ggTAAAAoHTW+/QJ0V3/n1urxvauAxcJ8nB0Dae2Gz94V2
KF7Eh48muATg8nmO2LijJ24zl1idXq94zqpqH/aJZHTJeYHFlIzWVoPf+ns6rCNaRI3bYZ
FT9ZoP/TId6fx9ENKM3cWCO3WdNwoB5/Y1CLAtnQKYh4XBtfiRiTJb4vkbHVoYO2OjEeVa
DYGd4itO0Tjy6+PmzCt38u0vHUrpFawaJK6A0=
-----END OPENSSH PRIVATE KEY-----
```

Conferido a Chave publica

```
C:\Users\França\.ssh>type id_Keyed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIIrD6PQCTK8gwUzduvOhDAx0hpa+TNIlxnCYYz9F1ggT sgtgladstone@gmail.com
```

Configuração do Servidor para receber a chave pública.

```
linux-server@linux-server-VirtualBox: ~
Firefox
Command 'cle?' not found, did you mean:
  command 'clex' from deb clex (4.6.patch8-1)
  command 'cleo' from deb cleo (0.004-4)
Try: sudo apt install <deb name>
linux-server@linux-server-VirtualBox:~$ sudo apt update
[sudo] password for linux-server:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,51
2 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [243
kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [
976 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [
219 kB]
Fetched 3,076 kB in 2s (1,684 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
44 packages can be upgraded. Run 'apt list --upgradable' to see them.
linux-server@linux-server-VirtualBox:~$
```

```
linux-server@linux-server-VirtualBox: ~
linux-server@linux-server-VirtualBox:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-instal
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
linux-server@linux-server-VirtualBox:~$ sudo systemctl start ssh
linux-server@linux-server-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-03-12 20:42:53 -03; 4s ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
Terminal
    Process: 8589 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 8590 (sshd)
      Tasks: 2 (limit: 4597)
     Memory: 2.4M (peak: 2.5M)
        CPU: 51ms
     CGroup: /system.slice/ssh.service
             ├─7389 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
             └─8590 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 12 20:42:52 linux-server-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell serve
Mar 12 20:42:52 linux-server-VirtualBox systemd[1]: ssh.service: Found left-over process 7389 (sshd) i
Mar 12 20:42:52 linux-server-VirtualBox systemd[1]: ssh.service: This usually indicates unclean termi
Mar 12 20:42:53 linux-server-VirtualBox sshd[8590]: Server listening on 0.0.0.0 port 22.
Mar 12 20:42:53 linux-server-VirtualBox sshd[8590]: Server listening on :: port 22.
Mar 12 20:42:53 linux-server-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

linux-server@linux-server-VirtualBox:~$
```

Definição da chave pública no Servidor.

```
        CPU: 51ms
     CGroup: /system.slice/ssh.service
             ├─7389 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
             └─8590 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 12 20:42:52 linux-server-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell serve
Mar 12 20:42:52 linux-server-VirtualBox systemd[1]: ssh.service: Found left-over process 7389 (sshd) in control group w
Mar 12 20:42:52 linux-server-VirtualBox systemd[1]: ssh.service: This usually indicates unclean termination of a previo
Mar 12 20:42:53 linux-server-VirtualBox sshd[8590]: Server listening on 0.0.0.0 port 22.
Mar 12 20:42:53 linux-server-VirtualBox sshd[8590]: Server listening on :: port 22.
Mar 12 20:42:53 linux-server-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

linux-server@linux-server-VirtualBox:~$ mkdir -p ~/.ssh
linux-server@linux-server-VirtualBox:~$ chmod 700 ~/.ssh
linux-server@linux-server-VirtualBox:~$ touch ~/.ssh
linux-server@linux-server-VirtualBox:~$ touch ~/.ssh/authorized_keys
linux-server@linux-server-VirtualBox:~$ chmod 600 ~/.ssh/authorized_keys
linux-server@linux-server-VirtualBox:~$ ls -la ~/.ssh
total 8
drwx------  2 linux-server linux-server 4096 Mar 12 20:46 .
drwxr-x--- 15 linux-server linux-server 4096 Mar 12 19:13 ..
-rw-------  1 linux-server linux-server    0 Mar 12 20:47 authorized_keys
linux-server@linux-server-VirtualBox:~$ nano ~/.ssh/authorized_keys
linux-server@linux-server-VirtualBox:~$ cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQDWZFKdv3x2TdrWnPceLy1a7rqaKljIE6dT6S5vg+CjY2FGPVK4xijVrRInDJ8Af
8zo8vDC/qv0xEyl02yi1JsQroNtcanbAWmtOy4OA8V9c0u4n/WjBEvo1kv4ZpqtbiODET+jI2k14DvFdlmFSXVumWrPieKhDYnrl
m9dM0+b+LoxRL/x8TQI6xVNWCpjRATE1AJPvjW/PQ/DoZyg9sql45y2yORHcBTaCgOxVgdAKffhGDtuI4MkfHuKnmm+ld4bAlRQAf
dqGIHUFAmdGLDfK2RZx/jfepzhORf7IqD60WMdNK1nJVzpr/UTI4QrKsOJ0SwXyf7MUOlzbPhZklQJEBRqlnFu3jQ4byW2PNH43hf
lGSbSEPdmPDe8T+iBsnaioCWMnTn8nOjDUuCD2r0PTfrhyL7TNOod3nbkrNxXR+uC70STljXeir1OefE0XQllgpTRuzm3K32nofo
sCVVJMk36eN/tudAUS/noun+ixTvJWpFOxzYGovB4eXNIR7ltPH6wjBNt5pWBk8z6vAbsfzrjw0AUXFJyQ3GLNUdxZhvgkABGvSk
Hw== aluno@DESKTOP-MEKIM56
linux-server@linux-server-VirtualBox:~$
```

Chamada do servidor na Máquina Cliente (TESTE DA CONEXÃO SSH)

```
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:awMeXoiI2rAynrAUi24rr8k69tHnS9G4zGfJ62m3Hkg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of known hosts.
Enter passphrase for key 'C:\Users\Aluno/.ssh/id_rsa':
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

linux-server@linux-server-VirtualBox:~$ whoami
linux-server
linux-server@linux-server-VirtualBox:~$
```